Primary Connection Bolts

The primary connection bolts include the bolts used for connecting the main frame members such as rafter, columns, mezzanine beams and joist, jack beams, strut tube and also including the bolts used for connecting the cold formed members, such as purlins, girts, eave struts, framed opening jambs & headers. The available standard stock as are grade 8.8 and diameter as listed in table 12.10. The procedure for different grades remains the same except for F_y and F_u values.

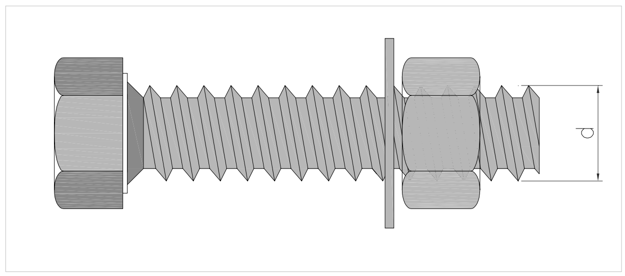


Table 12 .10 High Strength Bolts (ASD)

Diameter (d) (mm)	Gross Area (mm²) A_b	Single Shear (kN) R_{nv}	Double Shear (kN) 2 R_n	Tension (kN) R_{nt}
12	113.10	18.10	36.19	33.93
16	201.06	32.17	64.34	60.32
20	314.16	50.27	100.53	94.25
24	452.39	72.38	144.76	135.72
27	572.56	91.61	183.22	171.77
30	706.86	113.10	226.19	212.06
33	855.30	136.85	273.70	256.59
36	1017.88	162.86	325.72	305.36

All standard stock bolts are fully threaded and hence the shear values for 'threads in shear plane' should be considered in design.

Bracing Systems

A building is subjected to lateral forces in many direction due to wind, seismic, crane and other horizontal loads, for metal buildings these forces are resolved in two orthogonal directions. The main frame resist these lateral forces in the plane they are provided. In other direction a Bracings Systems is provided, these may consist of cross bracing or portal frames. These loads are eventually transferred to the foundations through a definite load path.

Following guide line should be followed when deciding upon the type of bracing and their configuration,

Wind and Seismic load Bracing in longitudinal direction

1. A maximum of 5 bays may be provided between braced bays.
2. A braced bay should not be located in the end bay of a building if the end wall system is a bearing frame end, unless it is required for many other reasons. In such cases where a bracing cannot be avoided in the end bays, the corner column may be either rotated or specially connection detail should be provided. The end wall columns should be checked for loading in all directions.
3. The bracing in roof and sidewall should be preferably in the same bays. If due to some reason such as framed openings etc, the sidewall bracing may be provided in one bay adjacent and proper load transfer should be designed and detailed to transfer the load through the strut member to the braced bay.
4. Roof bracing should be broken at the ridge line.
5. Cables / rods braces should preferably not exceed 15 m in length. If a cross bracing contains pieces longer than 15m, then the bracing may be broken to two sets of bracings with a strut member between them so that the rod/cable lengths is not exceeding 15 m.
6. Bracing of any one type or material, such as cable, rod, angle or portal bracing should be used in one wall. Mixing of different type of bracing, in one wall should be avoided.
7. It is preferable to use only one type of wall bracing in the whole building otherwise the lateral loads will have to be distributed as per the bracing stiffness.
8. Minimum diameter for cable bracing is 12 mm and 22 mm for rod bracing.
9. The maximum length of cable, rod or angle bracing should preferably not exceed 15 meters in length.
10. Use 75x75x6 double angles up to 12 meter and 100x100x8 double angles up to 15 meters, unless required otherwise by design.

Design of Eave Purlin

The design of eave purlin is similar to that of roof purlin, except for following,

- In addition to the loads in the major axis from roof, it is also subjected to some load in the minor axis from sidewall, hence it is necessary that, the distance between the last girt and eave point (500 mm) and also the distance between the eave point and second purlin (900 mm) is small, in order to minimize both the major and minor axis load on the eave purlin. Some portion of the vertical and horizontal loads is also transferred to the wall and roof sheeting respectively.

- In braced bays the eave purlin also acts as strut member in bracing truss for accumulated roof bracing loads and transfer these to sidewall bracing.

Eave purlin may be considered laterally restraint and fully supported by roof sheeting, and partially restrained by the wall sheeting through the eave angle connection. The wall sheeting is considered to provide full support for eave purlin against vertical deflections in the following cases,

- Fully sheeted walls, where the wall sheeting is resting on ground slab or beam
- Partially sheeted walls with block wall underneath where the bottom girt or base angle is resting on the block wall.

For portion of framed opening width, not exceed half the bay spacing, in fully sheeted walls, eave purlin may be considered as supported.

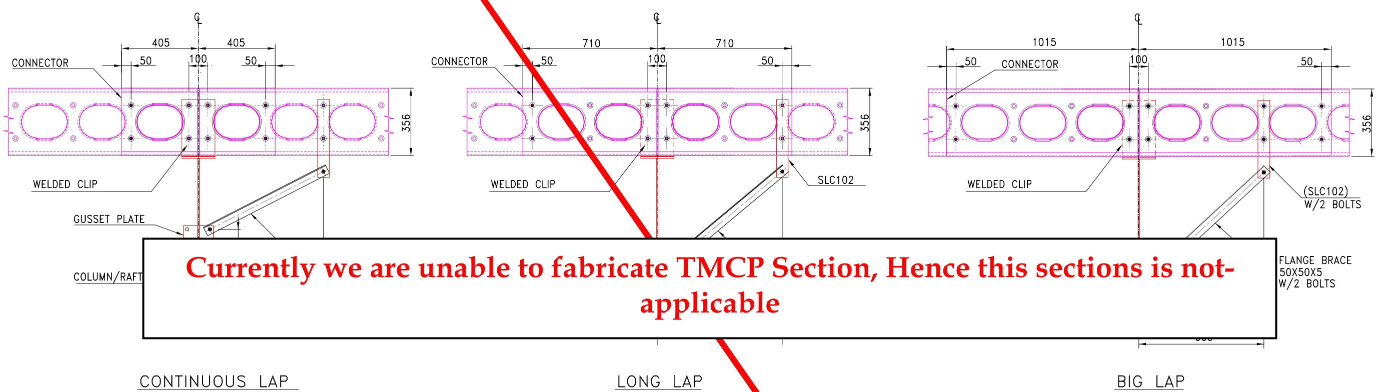
For other cases where the walls are partially sheeted from eave to some height above ground and open for access, adequate proper sag rod arrangement shall be provide with flayed sag rods at the top to carry the weight of the sheeting.

Section Properties and Capacities

The section properties and capacities of all available cold formed Sections are tabulated in the following tables. These may be calculated manually as per AISI Manual 2001 Section 3, the properties and capacities in the tables are calculated using RSG Software (CFS Version 4.14).

Connection Bolts

All bolts used for connection of purlin at the lap and to welded plate on the rafter / column are 16 mm diameter HSB Gr. 8.8. Grade of bolt is subject to change based on availability and management decision.



Currently we are unable to fabricate TMCP Section, Hence this sections is not-applicable

General

'M' sections of any dimension can be produced on the TMCP machine, with following limitations, but these will require coils of different widths.

Table 16.4 Section Profile on TMCP Machine

Dimension	Minimum (mm)	Maximum (mm)
Web depth	92	356
Flange width	41	76
Lip length	NA	25
Thickness	0.8	3.0
Minimum Length = 1930 mm		
NA - Denotes that the sections may be without lip (Open C or Stud)		

'M' Sections when used as Jambs and Headers for framed opening can be rolled with out the large holes.